# Local Credit, Global Settlement

## Enhancing Trust-Based Credit Creation with On-Chain Reserve Contracts

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# The Need for Monetary Expansion

## Blockchain Assets Are Inelastic

- Bitcoin, Ergo, and similar assets have **algorithmically predetermined emission schedules**
- Supply **cannot expand or contract** in response to real-world economic demand
- Entirely **disconnected** from productive activity, trade volumes, or credit needs

# ChainCash: On-Chain Notes with Collective Backing

- Peer-to-peer money creation through trust and/or collateral
- Notes backed by **all previous spenders** in the chain
- Redemption against **any reserve** in spending history,
  with re-redemption later possible against and earlier one
- Fully on-chain — **transparency and censorship resistance**

## Limitation

- Every transaction recorded on the blockchain
- Scalability and cost issues for frequent use
- Hard to apply in **community currency** settings with many small payments

# Offchain Payments Today: No Monetary Expansion

## Existing Bitcoin Layer-2 Systems

| System | Mechanism | Collateral Requirement |
|---|---|---|
| **Lightning Network** | Bidirectional payment channels | **100% locked on-chain** |
| **Cashu** | Chaumian blind signatures (e-cash) | **100% mint reserves** |
| **Fedimint** | Federated threshold custody | **100% federation backing** |

> **All three require full backing of off-chain liabilities with on-chain assets.**

## The Problem: No Credit Expansion

- Supply is capped by locked collateral
- Cannot respond to growing economic activity

### Contrast with Community Currencies

- Community currencies (LETS, Time Banks, WIR) thrive on **trust-based credit**
- Existing digital systems **discard this principle** by enforcing full collateralization
- **Trust within communities becomes an underutilized resource**

# Basis: A Protocol to unify Community Credit and Blockchain Assets

> **Enable elastic credit creation backed by trust, use on-chain reserves only when trust is insufficient**

## How It Works

- **Local trust-based IOUs** circulate within communities
- **On-chain reserves** serve as optional backing
- Individual IOU note acceptance
- **Off-chain payments** — low fees, no blockchain bottleneck
- **Tracker service** coordinates debt state transparently

```
┌──────────┐         IOU Note          ┌──────────┐
│  Payer   │  ──────────────────────►  │  Payee   │
│ (Issuer) │      (signed by both)     │(Receiver)│
└──────────┘                           └──────────┘
     │                                      │
     │                                      │
     └───────► Tracker (witness) ◄──────────┘
               (commits state to chain)
```

# Note Creation: Issuing Community Credit

1. **Payer** (e.g., a community member) creates an IOU note:
    - Payer's **signature**
    - Payee's **public key**
    - **Debt amount**
    - **Timestamp**
    - Optional: **reserve contract** for extra backing
2. **Payee** verifies payer's signature and note contents
    - **Acceptance is always voluntary** — each participant decides whom to trust
3. **Tracker signature** is obtained later
    - Required only for **redemption** against a reserve contract
    - Certifies the debt was recorded in tracker's authenticated state

# Note Redemption: From Trust to Settlement

## Standard Redemption (Trust + Collateral)

- Payee presents the **witnessed IOU** to the **reserve contract**
- Contract verifies payer's signature, tracker's signature, debt amount
- Collateral is **released to payee**

## Emergency Exit (Tracker Unavailable)

- After **3 days (2,160 blocks)**, tracker signature becomes **optional**
- Redeem using payer's signature + last committed tracker state
- **Replay attacks prevented** by timestamp verification

> This protects communities against tracker failure or censorship — funds are never locked indefinitely.

# Debt Transfer: Circulating Credit Within Communities

## Triangular Trade (No Blockchain Needed)

```
Before:                              After Transfer:
┌───────┐  owes 10     ┌───────┐    ┌───────┐  owes 5      ┌───────┐
│   A   │ ───────────► │   B   │    │   A   │ ───────────► │   B   │
└───────┘              └───────┘    └───────┘              └───────┘
                                       │ owes 5
                                       ▼
                                   ┌───────┐
                                   │   C   │
                                   └───────┘
```

- **B** requests **A** to co-sign:
  - Reduced debt A → B
  - New debt A → C
- Tracker witnesses both; old note replaced
- **A now owes C directly** — credit circulates!

# Tracker State: Transparent Debt Registry

## AVL Tree of Debt Relationships

$$\text{hash}(\text{payerKey} \,||\, \text{payeeKey}) \rightarrow (\text{totalDebt}, \text{timestamp})$$

- Cryptographically strong hash function
- Efficient lookups and **proof generation**
- Root digest **committed to blockchain periodically**
- Enables **emergency redemption** and **public verification**

## Governance Options

| Design | Trust Model | Best For |
|---|---|---|
| Single Server | Trusted operator | Small communities |
| Federation (MuSig2) | Threshold of signers | Municipal currencies |
| Sidechain (PoW) | Permissionless consensus | Regional / multi-community |

# Scenario 1: Rural Community Trading

## Disconnected Village with Occasional Connectivity

```
Village (no direct Internet)
┌──────┐    ┌──────┐    ┌──────┐
│ Alice│◄──►│  Bob │◄──►│ Carol│
└──────┘    └──────┘    └──────┘
     \          │         /
      \         │        /
       └────────┴───────┘
          Local Tracker
    (syncs via satellite/cellular)
```

- Trade on **credit** within community over **Bluetooth / LoRa**
- Tracker commits to blockchain **when Internet available**
- Redemption possible via **slow links (SMS, email)**
- **No forced collateralization** — trust is enough locally

# Scenario 2: Municipal Currency with Gradual Backing

## A City-Scale Complementary Currency

1. **Phase 1 — Pure Trust**: Local businesses issue IOUs to each other
    - No collateral required
    - Tracker (city federation) witnesses transactions
2. **Phase 2 — Selective Backing**: As trade expands to outsiders
    - Businesses optionally create **on-chain reserves**
    - Reserves back IOUs for **inter-city / low-trust** transactions
3. **Phase 3 — Hybrid Economy**: Local trust dominates internally, reserves handle external settlement

> **Monetary elasticity**: supply expands with community trust, contracts when trust is insufficient.

# Scenario 3: Solidarity Economy Network

## Cooperatives and Mutual Credit

- **Worker cooperatives** trade goods/services on credit
- **Mutual credit clearing** via debt transfer (triangular trade)
- **Tracker federation** run by cooperative assembly
- **Emergency exit** protects members against federation failure

```
Co-op A ──IOU──► Co-op B
   │                 │
   └──IOU──► Co-op C◄──IOU──┘

Circular debt cleared via tracker
No blockchain fees for day-to-day trade
```

# Security & Trust Minimization

## Tracker Cannot Steal Funds

- Tracker only **certifies inclusion** in state
- **Issuer's signature always required** for redemption
- Tracker cannot unilaterally create or redeem notes

## Anti-Censorship

- Pseudonymity allows picking new keys
- Reserve contract has **refund option** (long timeout)
- Future: blind signatures + zero-knowledge proofs

## Emergency Exit

- After timeout: tracker signature **optional**
- Same message format: `key || totalDebt || timestamp`
- Reserve owner's signature still required

# Why Blockchain? Why Ergo?

## Role of On-Chain Reserves

- **Not** for day-to-day transactions
- **Settlement layer** for low-trust or cross-community trades
- **Emergency anchor** if tracker fails
- **Limited blockchain use** — only where trust is truly insufficient

## Why Ergo?

- **UTXO model**: perfect for tokenized collateral
- **Smart contracts**: expressive redemption logic
- **Low fees**: viable for small settlements
- **PoW security**: no trusted validators
- **NiPoPoWs / SPV**: lightweight sync for intermittent connectivity

# Relation to Conference Themes

| Theme | How Basis Addresses It |
|---|---|
| **Digitalization & technologies** | Blockchain-backed IOUs, mesh network support, digital sovereignty |
| **Territories & participation** | Local credit for communities, voluntary acceptance, democratic tracker governance |
| **Public policies & government relations** | Municipal currency potential, no legal tender enforcement, individual choice |
| **Environment & sustainability** | Minimal energy use (off-chain payments), no mining for transactions |
| **History & theories** | Mutual credit tradition extended with cryptographic trust minimization |

# Contributions

1. **Basis protocol**: P2P credit creation with **optional** on-chain reserves
2. **Open-source implementations**: smart contracts + tracker server + clients
3. **Security analysis**: trust-minimization properties of tracker
4. **Real-world demos**: mesh network trading + solidarity economy scenarios

## Resources

- Paper & code: `github.com/ChainCashLabs/chaincash`
- Whitepaper: `docs/conf/conf.pdf`

# Thank You

## Local Credit, Global Settlement

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